

Trace

A small AI that reads a failed payment and works out what happened to the money — running on the bank's own machines, offline.

Oluwatomiwa Adeaga · deagatech

THE PROBLEM

₦1.07 quadrillion moved. Some of it got stuck.

EVERY TRANSFER IN NIGERIA, 2024

11.2 billion of them



₦1.07 quadrillion moved

₦19.12bn

refunded to customers

nearly double 2024

₦40.61bn

claimed in disputes, 2025

up from ₦17.13bn the year before

the cases someone must read

Every one of those disputes began as a case a person had to open, read and judge.

WHY IT IS URGENT

And the clock is set by the regulator

These are mandated ceilings, not internal targets.

Disputed PoS or web transaction



72 hours

Not-on-us ATM failure



48 hours

On-us ATM failure



24 hours

Failed airtime or data · Feb 2026



30 seconds

days → hours → seconds

manual reading does not scale down to seconds

The direction of travel is one way. In February 2026 the CBN and NCC set thirty seconds for failed airtime.

The answer is never in a database column

EXC-2026-0001930

N402,000.00

switch	91 · Beneficiary bank not available
ledger	DR 402,000.00 · NIBSS/REF957229178847
reversal	raised 20/08 06:15 · RV0765116324 acknowledged: true
settlement	no line for this session

remarks

19/08 17:15 BRANCH: informed cust escalation in progress
19/08 18:15 RECON: cust said she will come to branch tomorrow

what is missing

no credit leg anywhere on the account

The refund was raised.
The switch acknowledged it.

The money never
reached the customer.

A rule that checks
acknowledged == true
closes this case as settled.

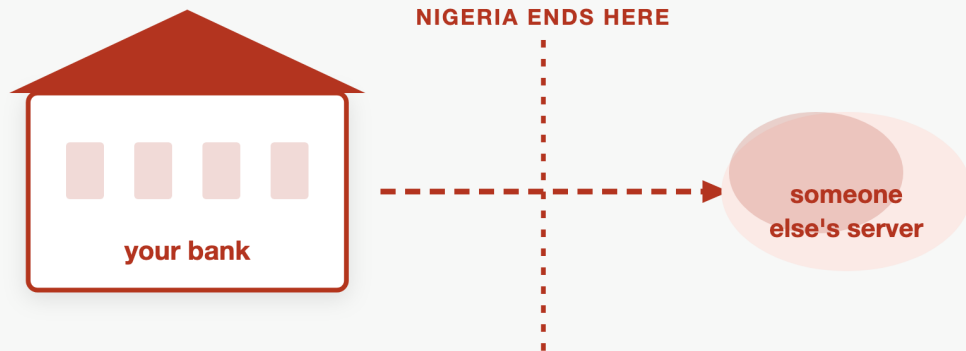
The evidence is an absence.
You have to notice what
is not there.

On this class a rules engine scores 46.4% and an untrained model 9.2%. Trained on the bank's own history: 73.1%.

THE CONSTRAINT EVERYONE HITS

So why not just use ChatGPT?

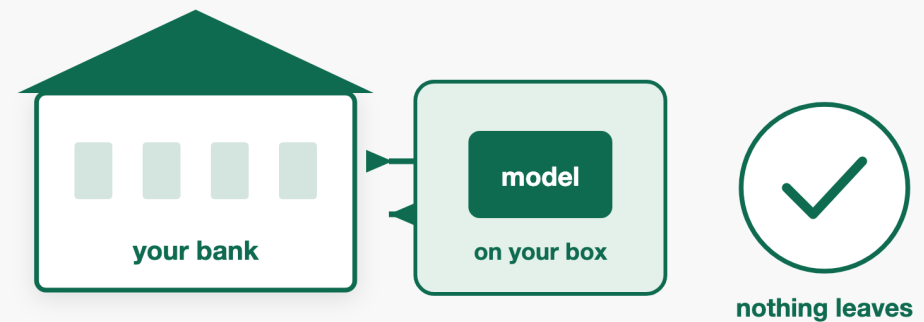
RENTING AN API



Customer names, account numbers and transaction history leave the country.

You own nothing. Price and access can change without you.

RUNNING YOUR OWN



One afternoon of GPU time to train.
Weights, data and test set you keep.

No per-call bill that grows with your own success.

For a Nigerian bank handling customer payment records this is not a preference. It is the difference between a product and a compliance problem.

What Trace is

A case goes in exactly as the desk receives it. An answer comes back that a person can check.

WHAT GOES IN

- the ledger entries
- the switch response
- the settlement file
- what the customer said
- the notes staff typed

One failed payment,
exactly as the desk gets it.



on the bank's own machine
offline



WHAT COMES BACK

- **What went wrong**
one of seven kinds of break
- **Where the money went**
written out, step by step
- **What to do about it**
reverse, credit, recall, close, escalate
- **How sure it is**
so the doubtful ones can be found
- **Whether a person must see it**
and it can always say yes

An answer a person can check, not just a label.

The evidence is an absence

It reads the free text and the ledger together, so it can notice what is not there.

The data cannot leave

It is small enough to run on the bank's own hardware. Nothing is sent anywhere.

The regulator's clock

It answers in about a second, so the deadline stops being the binding constraint.

THE RESULT

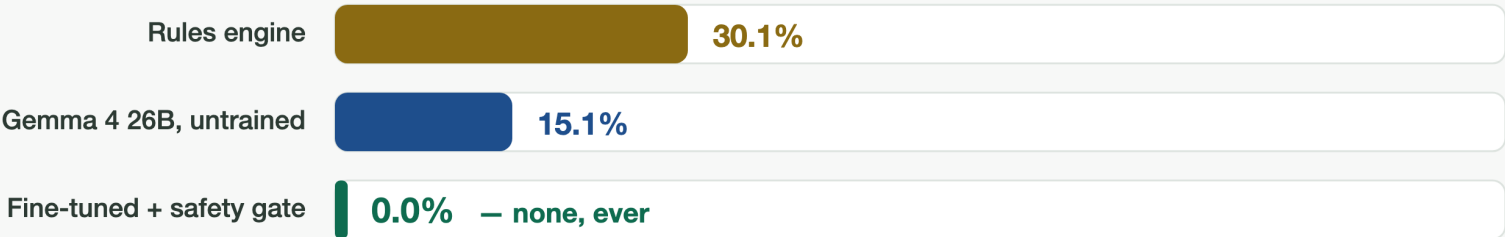
A 2B model beats a 26B model that never saw your data

GETTING THE PROBLEM RIGHT



FRAUD CLOSED WITH NOBODY LOOKING

lower is better

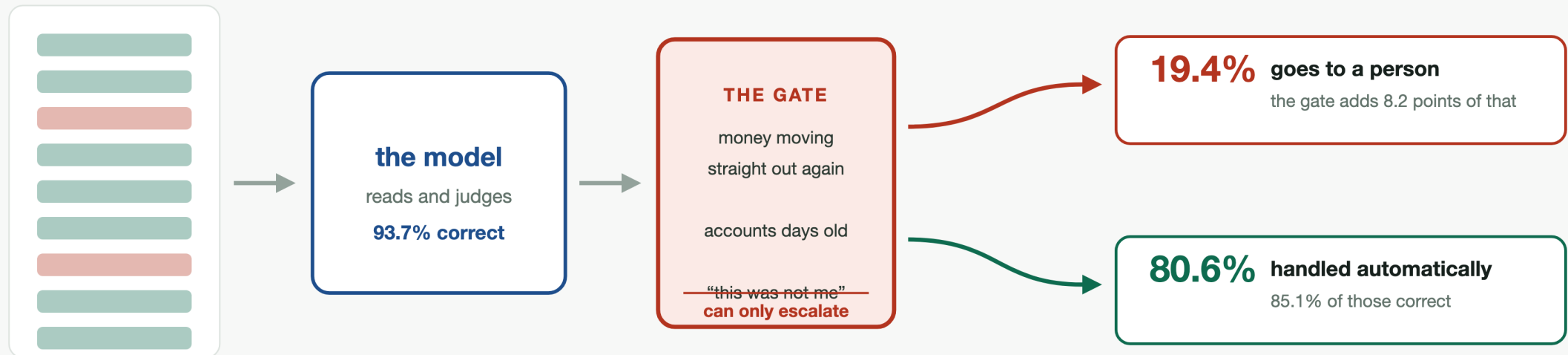


1,000 held-out cases, none seen during training. Every prediction published at github.com/adg-all-day/payment-exceptions-triage

THE SAFETY GATE

It can never quietly close a fraud

EVERY CASE



The model may be wrong. The system may not be dangerous.

Fraud closed with nobody looking: 30.1% under a rules engine, 0.0% here.

WHAT IT TAKES

Built in fourteen hours for about ₦4,200

Not a research project. One rented graphics card, one afternoon of training, and a test set built before the model so the numbers could not flatter it.

Training the model	2 h 42 m	one rented NVIDIA L4
Four evaluation runs	1 h 40 m	5,000 cases scored
Total compute spend	≈ ₦4,200	\$3 at ₦1,399/\$, then deleted
The finished model	185 MB	the adapter, measured on disk

A bank's exception desk costs more than this in a morning.

What this is not, yet

Everything above is measured on invented data. Here is precisely what has not been proved.

It has not seen one real transaction

Every case is synthetic, generated so the correct answer is certain. That makes the comparison honest, but it does not prove it survives your data.

It has not been tested against drift

The measurement that matters is training on one period and testing on a later one. That needs real history with dates on it.

One class is still weak

Refund-raised-but-never-paid sits at 73%, well below the rest. It is the obvious next thing to fix.

Three datasets were thrown away

Once the clues were too tidy and rules scored 96.6%. Once the evidence was deleted. Once the answers cited evidence that was not there. All recorded in the public build log.

Most pitches show you the wins. The failures are the reason to believe the wins.

DEAGATECH

The ask

Not prize money.

Anonymised history

Six months of closed exceptions from one break type.
Account numbers hashed, names removed — the text is what matters.

Nothing identifying ever leaves your building.

One person, six weeks

A named contact in operations who can tell us when an answer is wrong and why.

This is the part that cannot be bought.

A number to beat

What your desk resolves per analyst per day today, and how often a case is reopened.

We will publish whether we beat it or not.

You keep the model, the weights, and the test set. If you never speak to us again, it still runs.